



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Computational Fluid Dynamics

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

Fritz Ali Agildere

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Basic Equations

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Finite Volume Method

Finite Element Method

Practical Implementation

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Rayleigh-Taylor Instability

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Fluid Elements

- *Fluids*: things that flow, conventionally liquids and gases, contrast solids with rigid lattice
- *Elements*: region with definable local variables, fulfilling certain conditions
 - small enough to ignore systematic variations, size below any characteristic scale length

$$\ell \ll \left| \frac{q}{\nabla q} \right|$$

- large enough to ignore discreteness noise, statistically sufficient number of particles

$$n\ell^3 \gg 1$$

- in collisional fluids, large enough to communicate local conditions, requires interactions

$$\ell \gg \lambda$$

Describing Fluid Equations

- *Eulerian View*: cells of small volumes in fixed grid, flow is defined for all $\mathbf{t}$ at position $\mathbf{r}$ with rate of change $\partial/\partial\mathbf{t}$ for any measurable quantity Q
- *Lagrangian View*: follow particular fluid elements, comoving reference system with flow evaluated in each one via $D/D\mathbf{t}$ as the rate of change
- *Translating Relation*: add convective derivative term to account for movement of fluid element

$$\frac{DQ}{Dt} = \frac{\partial Q}{\partial t} + \mathbf{u} \cdot \nabla Q$$

Continuous Variables

■ *Primitive Fields:*

- velocity $\mathbf{u}$ (vector)
- density ρ (scalar)
- pressure p (scalar)
- temperature T (scalar)

■ *Conserved Quantities:*

- mass M (scalar)
- energy E (scalar)
- momentum $\mathbf{P}$ (vector)

■ *Derived Values:*

- viscosity η (scalar)
- vorticity $\boldsymbol{\omega}$ (vector)

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Dimensionless Numbers

- *Reynolds Number*: ratio of inertial to viscous forces

$$R = \frac{\rho u L}{\eta} = \frac{u L}{\nu}$$

- $R < 1000$: Laminar Flow (reversible and ordered stream)
 - ▶ $R < 1$: Stokes Regime (without any separation)
 - ▶ $10 \lesssim R \lesssim 40$: Separation Zone (stable and stationary vortices)
- $R > 1000$: Turbulent Flow (stochastic and chaotic stream)
 - ▶ $40 \lesssim R \lesssim 4000$: Transitional Kármán Street (vortex shedding)
 - ▶ $R > 4000$: Kolmogorov Energy Cascade (fully developed phase)

Dimensionless Numbers

- *Mach Number*: ratio of bulk velocity magnitude to local sound speed

$$M = \frac{u}{c}$$

- $M \lesssim 0.3$: Incompressible (negligible density variations, pressure signals faster than flow)
- $M \approx 1$: Transonic (local excess of sound speed, weak shocks create drag)
- $M > 1$: Supersonic (thin front of discontinuity, bow shock heating)
- $M > 3$: Hypersonic (ram pressure stripping, dissociation and ionization)

Bulk Transport

- *Continuity Equation*: density ρ must increase where a fluid is compressed

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0$$

- *Linear Advection*: velocity field $\mathbf{v}$ is free of divergence for incompressible fluids

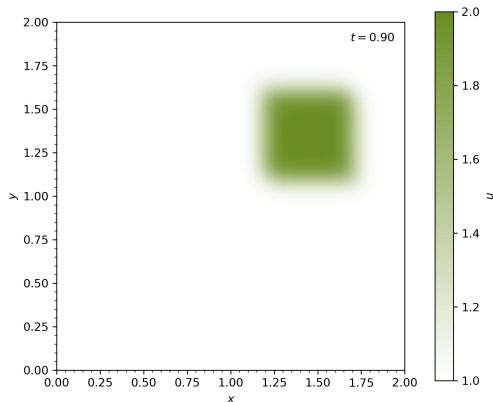
$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{u} = 0$$

- *Nonlinear Advection*: velocity field $\mathbf{u}$ is carried by itself instead of fixed background flow

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = 0$$

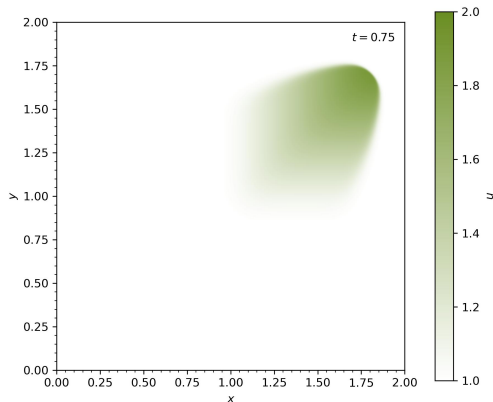
Linear Advection

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Lower:* constant value
 - *Upper:* unconstrained
- *Observation:*
 - fixed profile
 - linear translation



Nonlinear Advection

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Lower:* constant value
 - *Upper:* unconstrained
- *Observation:*
 - shock formation
 - rarefaction tail



Stochastic Transport

- *Diffusion*: kinematic viscosity ν describes spread due to random motion

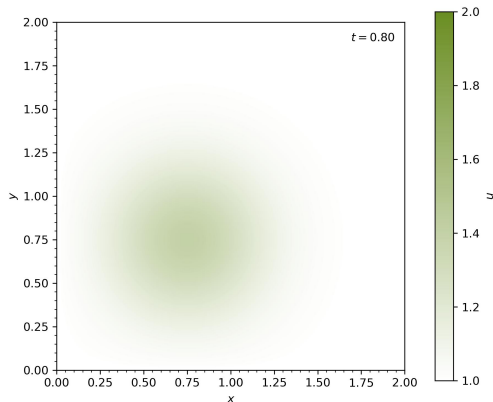
$$\frac{\partial u}{\partial t} = \nu \nabla^2 u$$

- *Bateman-Burgers Equation*: combines shock generation with counteracting diffusive term

$$\frac{\partial u}{\partial t} + (u \cdot \nabla)u = \nu \nabla^2 u$$

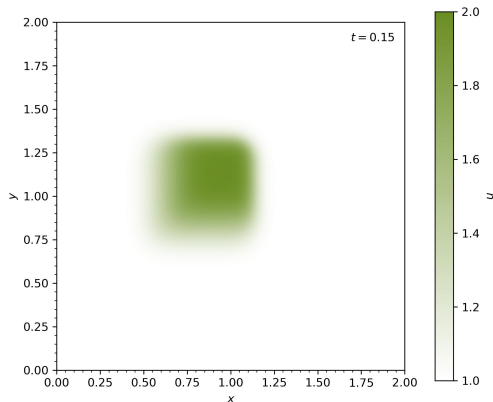
Diffusion

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Both:* constant value
- *Observation:*
 - stationary decay
 - edge smoothing
 - spreading out



Bateman-Burgers Equation

- *Initialization:*
 - square wave
- *Boundaries:*
 - *Both:* constant value
- *Observation:*
 - nonlinear steepening
 - viscous smoothing
 - dissipating translation



Equilibrium Search

- *Laplace Equation*: maximally smooth solution dictated solely by boundary conditions

$$\nabla^2 p = 0$$

- *Poisson Equation*: extension for steady state to accomodate internal sources and sinks

$$\nabla^2 p = f(r)$$

- Gravity: $\nabla^2 \Phi = 4\pi G\rho$
- Electrostatics: $\nabla^2 V = -\rho/\epsilon$
- Pressure: $\nabla^2 p = \rho \nabla \cdot ((u \cdot \nabla)u)$

Laplace Equation

- *Initialization:*

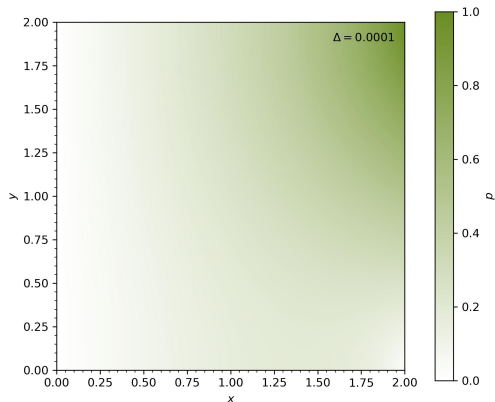
- empty field

- *Boundaries:*

- *Left:* fixed zero
- *Right:* linear slope
- *Top & Bottom:* no gradient

- *Observation:*

- iterative diffusion
- smooth relaxation



Poisson Equation

- *Initialization:*

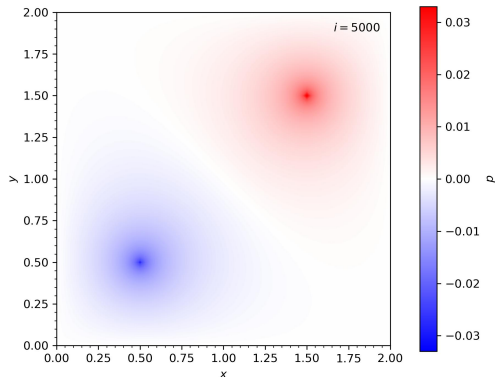
- empty field
- positive source
- negative sink

- *Boundaries:*

- All: fixed zero

- *Observation:*

- dipole field potential
- converging relaxation



Navier-Stokes Equations

- *Momentum Equation*: ensure $\nabla \cdot \mathbf{u} = 0$ by including ∇p to instantaneously cancel out any divergence

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\frac{1}{\rho} \nabla p + \nu \nabla^2 \mathbf{u}$$

- *Pressure Poisson*: require p to have relaxed to its equilibrium for all t before projecting

$$\nabla^2 p = \rho \nabla \cdot ((\mathbf{u} \cdot \nabla) \mathbf{u})$$

- fulfill at each step to accurately evolve initial configuration under given boundary conditions

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Lid Driven Cavity Flow

■ Initialization:

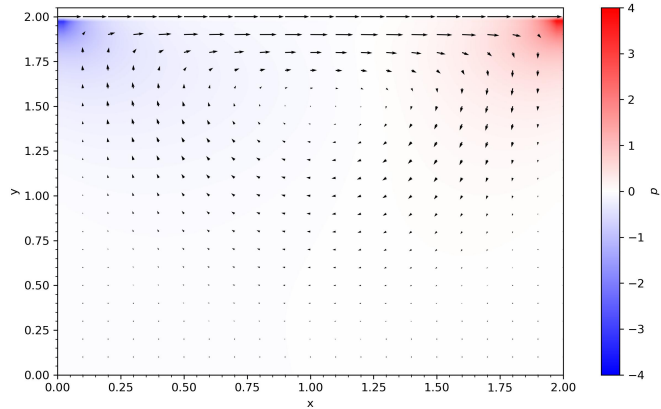
- vanishing velocity
- zero pressure

■ Boundaries:

- Top: $u = 1$ & $\nabla p = 0$
- Bottom: $u = 0$ & $p = 0$
- Left & Right: $u = 0$ & $\nabla p = 0$

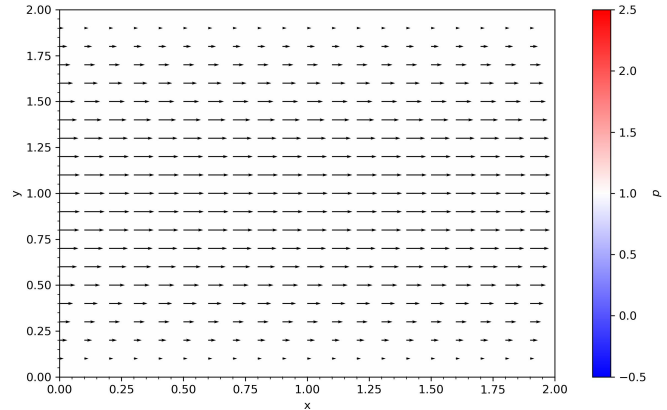
■ Observation:

- vortex formation
- pressure gradient



Pressure Driven Channel Flow

- *Initialization:*
 - vanishing velocity
 - unity pressure
 - *Poiseuille*: constant source
- *Boundaries:*
 - *Top & Bottom*: $u = 0$ & $\nabla p = 0$
 - *Left & Right*: periodic
- *Observation:*
 - curved velocity profile
 - boundary layer formation



Locally Forced Channel Flow

■ Initialization:

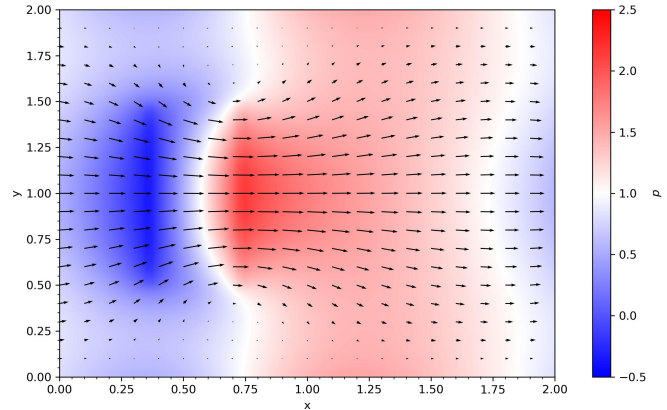
- vanishing velocity
- unity pressure
- rectangular source

■ Boundaries:

- Top & Bottom: $u = 0$ & $\nabla p = 0$
- Left & Right: periodic

■ Observation:

- central jet formation
- recirculation eddies



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Conceptual Overview

- Idea: pointwise derivative estimation with series expansion via differences between adjacent nodes
- Advantages:
 - easy implementation
 - computationally efficient
 - highly accurate for simple problems
- Drawbacks:
 - struggles with complex geometries
 - strong numerical diffusion at low orders
 - inflexible rectilinear grid with fixed resolution
 - mass and energy can leak due to numerical errors

Integration Techniques

■ *Spatial:*

- *Downwind Scheme:* looks forward in direction of transport, unstable due to causality violation
- *Upwind Scheme:* looks backward against the direction of transport, very stable but diffusive
 - ▶ used to model transport with defined direction
- *Central Difference:* considers both directions, more precise but can lead to oscillations
 - ▶ used to model symmetric dispersion processes
- *Multi Point Stencil:* higher order central method for computing ∇^2 in multiple dimensions

■ *Temporal:*

- *Backward Euler:* looks to future to calculate next step implicitly by inversion, extremely stable
- *Forward Euler:* looks to past to find next step explicitly, simple but unstable for large intervals
- *Leapfrog:* reaches back to previous step instead of current one, symplectic and invertible

Integration Techniques

■ *Advanced:*

- *Lax-Wendroff*: take intermediate step, equivalent to series expansion to second order
- *Runge-Kutta*: take trial steps to estimate curvature, then make informed jump
- *Predictor-Corrector*: take roughly predicted step, then recalculate with new information
- *Leapfrog*: preserve phasespace volume and total energy, enable backward reconstruction

■ *Courant-Friedrichs-Lewy Condition*: $\frac{v\Delta t}{\Delta x} \leq 1$

■ *Pure Diffusion Condition*: $\frac{2v\Delta t}{\Delta x^2} \leq 1$

Discretized Formulation

- *Convection:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = 0 \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = 0$
- *Diffusion:* $\frac{\partial u}{\partial t} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$
- *Bateman-Burgers Equation:* $\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} = v \frac{\partial^2 u}{\partial x^2} \longrightarrow \frac{u_i^{n+1} - u_i^n}{\Delta t} + u_i^n \frac{u_i^n - u_{i-1}^n}{\Delta x} = v \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{\Delta x^2}$
- *Laplace Equation:* $\frac{\partial^2 p}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \longrightarrow \frac{p_{i+1,j}^n - 2u_{i,j}^n + u_{i-1,j}^n}{\Delta x^2} + \frac{u_{i,j+1}^n - 2u_{i,j}^n + u_{i,j-1}^n}{\Delta y^2} = 0$

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- Idea: calculate flux through faces of control volumes that divide the domain using integral form

$$\frac{\partial}{\partial t} \int_V Q dV + \oint_S (F \cdot n) dS = \int_V \Sigma dV$$

- Advantages:
 - inherently strictly conservative for total mass and energy
 - unstructured grid handles complex geometries well
 - easily deals with shock discontinuities
- Drawbacks:
 - mathematically complex at higher order accuracy

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- Idea: optimize polynomial shape function to approximate solution across whole grid elements
- Advantages:
 - highly mathematically robust
 - precise gradient calculation at boundaries
 - handles extremely complex mesh geometries well
- Drawbacks:
 - computationally expensive and memory intensive
 - bad stability for fast turbulent flows

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Setup

■ Initialization:

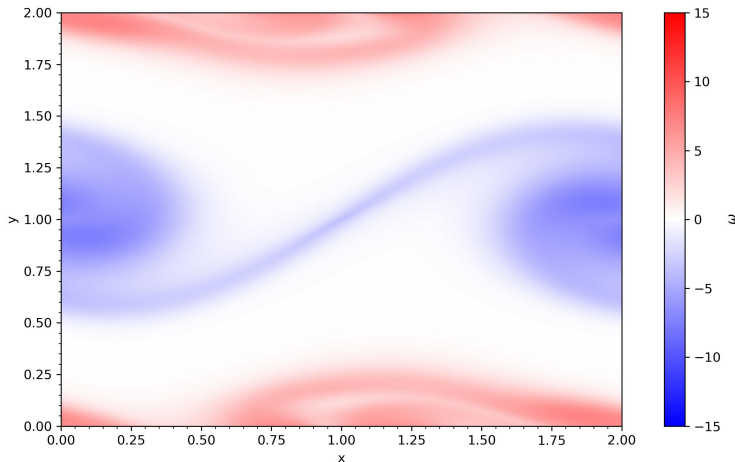
- oppositely directed horizontal velocity layers
- thin shear boundary modeled with $\tanh(y)$ for smooth transition
- vertical perturbation using $\sin(x)$ to trigger instability
- scalar curl $\omega = \partial v / \partial x - \partial u / \partial y$ as visualization

■ Boundaries: fully periodic for effectively toroidal domain

■ Observation:

- shear layer rolls up into distinct spiral structures
- vortices grow into rotating billows
- mixing occurs and creates turbulence

Visualization



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Setup

■ Initialization:

- fluid stratification with vertically resting layers
- thin interface modeled with $\tanh(x)$ for smooth transition
- horizontally perturbed using $\sin(y)$ to trigger instability
- concentration C to calculate coupled buoyancy forces F and as visualization
- Boussinesq: density approximated to be the same in both fluids

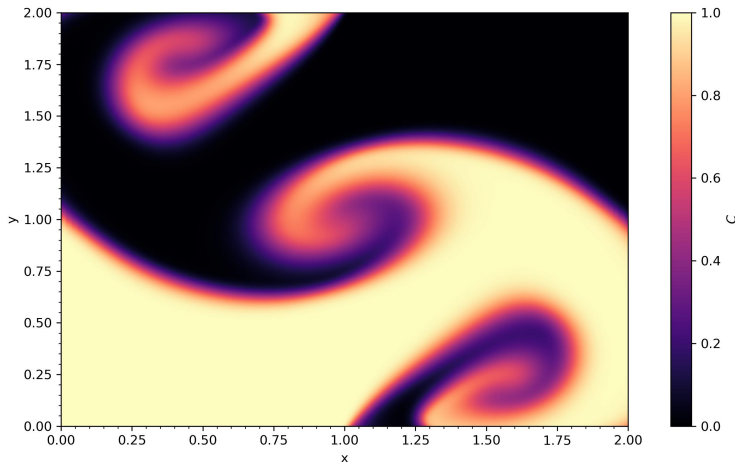
■ Boundaries:

- Top & Bottom: periodic for an infinitely repeating domain
- Left & Right: fixed values to contain the fluids inside plot

■ Observation:

- fingers extend from instability and form mushroom caps
- Kelvin-Helmholtz Instabilities: spiraling vortices from friction lead to turbulent mixing

Visualization



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■ Frameworks:

- *Smoothed Particle Hydrodynamics*: mesh free interactions based on kernel weighting
- *Lattice Boltzmann Method*: cells simulate statistical particle collisions and streaming
- *Adaptive Mesh Refinement*: grid resolution increases where higher precision is required

■ Regimes:

- | | |
|--|---|
| ■ <i>Plasma</i> : use magnetohydrodynamics | ■ <i>Degeneracy</i> : need quantum effects |
| ■ <i>Relativity</i> : use relativistic hydrodynamics | ■ <i>Collisionless</i> : need initial distributions |

■ Applications:

- | | | |
|---------------------|-------------------|--------------------|
| ■ stellar evolution | ■ accretion disks | ■ cosmic filaments |
|---------------------|-------------------|--------------------|

Thank You!

- C. Clarke & B. Carswell, *Principles of Astrophysical Fluid Dynamics* (Cambridge University Press) 2007
- L. A. Barba, *CFD Python: 12 Steps to Navier-Stokes*¹ 2022
- P. Mocz, *Medium*² 2020-2025

Resources on GitHub: [here](#)